

Optimization of Reactive Ink Formulation for Controlled Additive Manufacturing of Copolymer Membrane Functionalization

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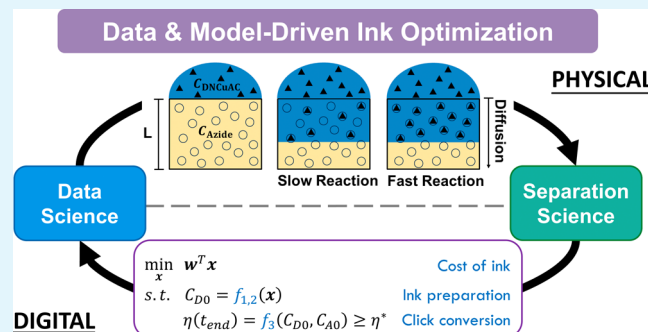
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ABSTRACT: Multifunctional, nanostructured membranes hold immense promise for overcoming permeability–selectivity trade-offs and enhancing membrane durability in challenging molecule separations. Following the fabrication of copolymer membranes, additive manufacturing technologies can introduce reactive inks onto substrates to modify pore wall chemistries. However, large-scale implementation is hindered by a lack of systematic optimization. This study addresses this challenge by elucidating the membrane functionalization mechanisms and optimal manufacturing conditions using a copper(I)-catalyzed azide–alkyne cycloaddition (CuAAC) “click” reaction. Leveraging a data science toolkit (e.g., nonlinear regression, uncertainty quantification, identifiability analyses, model selection, and design of experiments), we developed two mathematical models: (1) algebraic equations to predict equilibrium concentrations after preparing reactive inks by mixing copper sulfate, ascorbic acid (AA), and an alkynyl-terminated reactant; and (2) reaction–diffusion partial differential equations (PDEs) to describe the functionalization process. The ink preparation chemistry with side reactions was validated through pH and UV–vis measurements, while the diffusion and kinetic parameters in the PDE model were calibrated using time-series conversion of the azide moieties inferred from Fourier-transform infrared spectroscopy. This modeling framework avoids redundant experimental efforts and offers a functionalization protocol for scaling up designs. Ink optimization problems were proposed to reduce the use of expensive and environmentally insulting ink materials, i.e., Cu(II), while ensuring the desired chemical distributions. With optimal ink formulation Cu(II)/AA/alkyne = 1:1:2 identified, we uncovered trade-offs between Cu(II) usage and functionalization time; for example, in continuous roll-to-roll manufacturing with a conserved functionalization bath setup, our optimal operational conditions to achieve $\geq 90\%$ functionalization enable at least a 20% reduction in total copper investment compared to previous experimental results. The data science-enabled ink optimization framework is extendable for on-demand multifunctional membranes in numerous future applications such as metal recovery from wastewater and brine.

KEYWORDS: mathematical modeling, CuAAC click reaction, UV–visible spectroscopy, Fourier-transform infrared spectroscopy, data science, digital twin



1. INTRODUCTION

Nanofiltration (NF) membranes present a pivotal frontier in membrane science, with their unique selective rejection between monovalent and multivalent ions,¹ shaping their applicability across various fields. For example, in the realm of water treatment, NF membranes have been used for water softening (calcium and magnesium removal),² sulfate removal,³ and heavy metal removal.⁴ However, state-of-the-art commercially available NF membranes suffer from trade-offs between permeability and selectivity due to their nanostructure determining size-selective transport mechanisms.⁵ Additionally, deleterious fouling, which diminishes the treatment capacity and lifetime of NF membranes, remains another major concern.⁶ Addressing these issues is paramount for harnessing NF membranes in separations critical to global grand

challenges such as equitable access to clean water, affordable sustainable energy, and circular economies.^{7,8}

Efforts to develop self-assembled materials have reached a plateau in controlling pore size and distribution and have delivered high recovery and throughput in size-selective separations.^{9–11} Further advancements in membrane separation now focus on separating species based on chemical identity. For instance, introducing charged functionalities to

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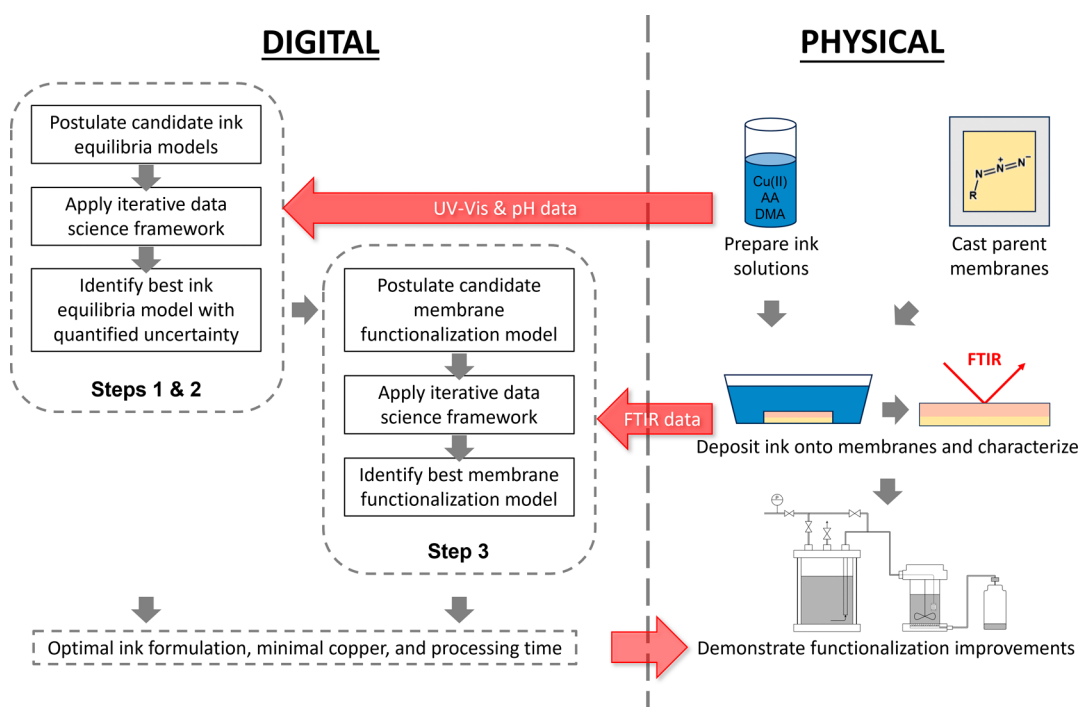


Figure 1. Digital (left) and physical (right) workflows are combined to model and optimize reactive ink solutions and processes for membrane functionalization. Steps 1 to 3 correspond to reaction steps in Figure 2.

membrane pore walls equips NF membranes to electrostatically reject ions without sacrificing permeability, demonstrating great potential in selective separation beyond size filtering (i.e., separation of similar-sized molecules).^{12,13} Other beneficial characteristics related to membrane performance can also be introduced by modifying the pore wall chemistry. For example, antifouling characteristics can be enhanced with the inclusion of hydrophilic and charged moieties, e.g., zwitterionic moieties,^{14,15} on the membrane surface layer. To this end, membranes capable of integrating multiple functionalities in a controlled manner will enable a fit-for-purpose paradigm¹⁶ for the on-demand design of membrane processes.

Self-assembled materials, providing a precise and fixed nanostructure, serve as an excellent platform for fabricating multifunctional membranes.¹⁷ Postfabrication functionalization strategies can be integrated with additive manufacturing (e.g., 3D printing) to create novel membrane structures with tailored chemistry distributions. In this process, reactive solutions (i.e., inks) are deposited onto the membrane, gradually penetrating and modifying the pore wall structure. Surface functionalities can be patterned using inkjet printing,¹⁸ while varied functionalities across the membrane depth (i.e., Janus membranes) can be achieved through sequential casting.¹⁹ Controlling the distribution of moieties along multiple axes requires delicate control over ink formulations, sequencing, and timing of exposures between the membrane and reactive solutions to achieve a well-designed chemistry architecture, coupling optimal designed performance in target applications.²⁰ For example, Hoffman et al.^{16,21} demonstrated that fast kinetics that keep the membrane functionalization process in the transport-limited regime (i.e., where the reaction rate is much faster than the diffusion of species) ensure discrete chemical patterns. The adoption of roll-to-roll manufacturing allows for the integration of membrane fabrication and postfabrication functionalization processes with enhanced

scalability and efficiency.^{16,22} Large-scale manufacturing necessitates consistent polymer casting-to-coating speeds and space to provide sufficient functionalization and rinsing time.²¹ Additionally, commercializing multifunctional membranes requires reducing the cost of ink materials and avoiding chemical waste.²³ These considerations underscore the challenges of postfabrication functionalization in achieving desired membrane characteristics targeted to fit-for-purpose separation processes.²⁴

“Click” chemistries (e.g., copper-catalyzed azide alkyne cycloaddition, thiol–ene), with their precise and high-yield molecular binding under mild conditions, offer facile, versatile, and scalable reaction pathways whose time scales are comparable to roll-to-roll techniques.²⁵ Understanding the intricacies of click reactions is crucial for modulating the membrane chemistry effectively. The copper(I)-catalyzed azide–alkyne cycloaddition (CuAAC) reaction is a prominent example that converts azide moieties to triazole derivatives, which has been utilized to introduce positive charge, negative charge functionalities, and zwitterionic moieties to membranes.^{19,26,27} Studies elucidating the mechanism of CuAAC suggest that two Cu(I) atoms are involved as a dinuclear copper–alkyne π complex (DNCuAC) in the cycloaddition steps.²⁸ However, equilibrium processes forming the productive species subject to multiple intermediaries hinder the estimation of DNCuAC concentration in deciding the cycloaddition rate directly. Previous experimental efforts to determine “click” equilibrium constants through ion-current analysis showed an order of magnitude difference^{29,30} and are too resource-intensive to consider the extension in membrane functionalization studies using different alkyne-terminated reactants. A systematic calibration of the reactions tailored in membrane functionalization with reduced experimental effort will facilitate optimal design (e.g., minimize cost and environmental impact) of the large-scale manufacturing

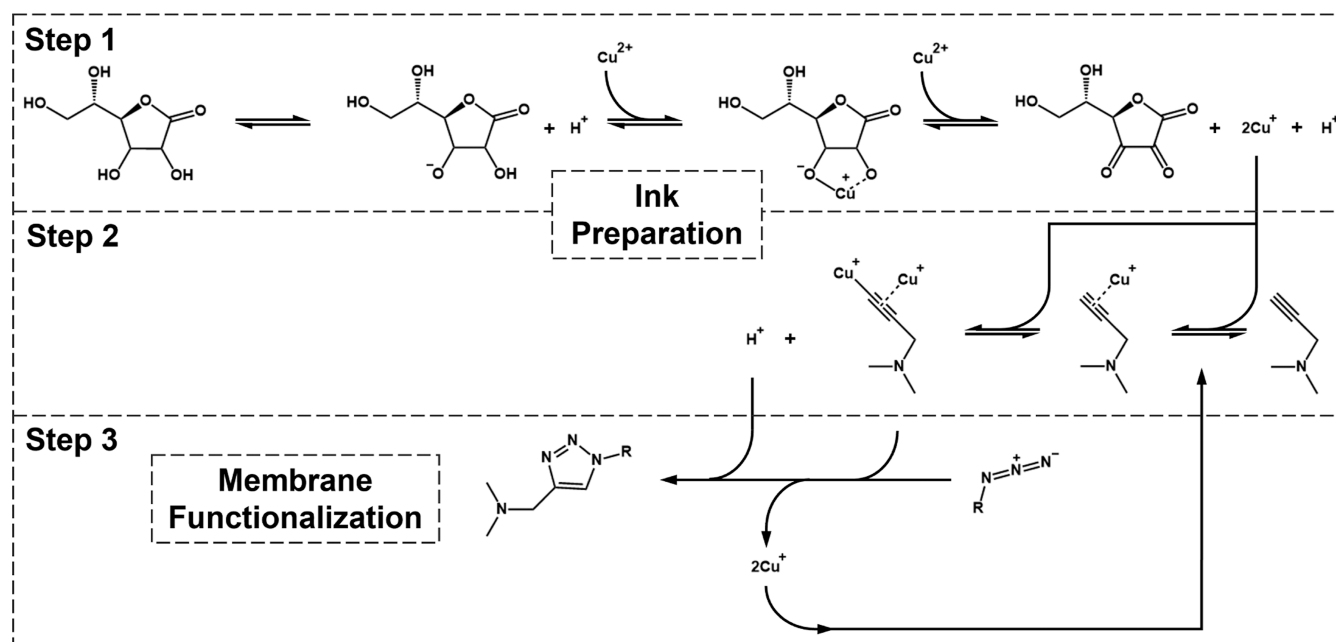


Figure 2. Schematic of the CuAAC reaction network. In step 1, Cu^{2+} is converted to Cu^+ by ascorbic acid (AA). In step 2, a mononuclear copper π complex (MCuPIC) and then a dinuclear copper-alkyne π complex (DNCuAC) are formed from Cu^+ and alkyne-terminated reactant. Step 3 is the “click” of the alkyne to the azide. Steps 1 and 2 correspond to the ink preparation stage, and step 3 represents the membrane functionalization process.

process. In this regard, dynamic functionalization progress monitored through Fourier-transform infrared (FTIR) spectroscopy facilitates modeling and verification of “click” progress,²¹ ensuring meticulous control over the functionalization process.

Computational tools and data analytics are becoming indispensable in unraveling the complexities in membrane science.^{7,31,32} For instance, leveraging dynamic data and nonlinear parameter estimation, diafiltration experiments for statistical membrane characterization provide coupled experimental and analytical tools capable of exploring the impact of different chemical pattern thicknesses or widths on membrane performance.³³ Model-based design of experiments (MBDoe) techniques contribute valuable insights into experimental design for scientific modeling with minimal experimental time and resource investment.³⁴ MBDoe frameworks start by elucidating optimal experimental conditions needed to discriminate between alternative model candidates, identifying most suitable reaction pathways or intermediates representing the system.³⁵ Next, MBDoe provides quantitative metrics interpreting parameter estimability with local sensitivities^{36,37} and allows minimizing of experimental trials/time required for statistically reliable estimation of model characteristic parameters^{38,39} in reactive systems. Recently, MBDoe was extended to identify trade-offs between information gain and cost for data collection,⁴⁰ which can be used to select cost-efficient experimental techniques for membrane functionalization process modeling. Extensive modeling capabilities and well-implemented numerical methods within Python optimization modeling objects (Pyomo) software package^{41,42} enhance the potential to explore temporal and spatial chemical distribution and to simulate and analyze the intricate functionalization process, thereby providing a deeper understanding of multifunctional membrane manufacturing.

This study seeks to develop cost-efficient and environmentally responsible CuAAC-based functionalization processes

in membrane manufacturing by integrating data science methods and experimental techniques. We present a comprehensive data science workflow to mathematically elucidate how additive manufacturing materials preparation, i.e., reactive ink formulation, affects the functionalization progress. Our exploration encompasses algebraic equations to predict the equilibria of the ink preparation, followed by partial differential equations (PDEs) that couple kinetics and transport to describe functionalization progress in the membrane. Figure 1 outlines the main steps and interactions between digital and physical investigations in this work. Section 2 describes the mathematical models, and for brevity, only essential models are discussed, while additional candidate models are provided in the Supporting Information. Section 3 describes the physical and digital (e.g., data science and computational) experimental methods. This integrated approach is not limited to the CuAAC reaction mechanism but rather provides a systematic framework to identify accurate, predictive, and informative models for membrane functionalization, delivering optimal membrane manufacturing design criteria for both continuous and precise production of multifunctional membranes in general.

2. THEORY AND MATHEMATICAL MODELS

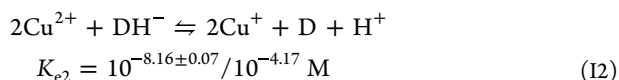
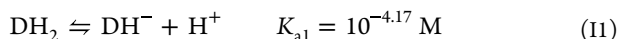
The reactive ink optimization problem is built upon two stages in the postfabrication functionalization process, as shown in Figure 1. First, the reactive ink solution is prepared; the equilibrium of the ink species is monitored through the pH and UV–vis for chromogenic components. Subsequently, functional groups are introduced by depositing ink solution on the membrane surface, where the transition of the parent azide group is monitored by FTIR. This section presents mathematical models used to optimize the ink formulation and the membrane functionalization process.

2.1. CuAAC Reaction Network. The CuAAC reaction network refers to a plethora of possible equilibrium reactions of copper in the ink solution and the catalytic azide–alkyne cycloaddition on the membrane. Figure 2 shows the three pivotal steps of the CuAAC reaction network. Step 1 reduces Cu^{2+} ions into catalytically active Cu^+ by ascorbic acid (AA).⁴³ Step 2 forms the DNCuAC. According to the prevailing literature,⁴⁴ a reversible reaction between the alkyne-terminated reactant and Cu^+ leads to the formation of a Cu(I) –alkyne complex. Recent investigations believe that DNCuAC rather than the mononuclear copper π complex (MCuPIC) participates in the final step of CuAAC.²⁸ Finally, step 3 couples the azide moieties on the membrane and the DNCuAC, which completes the “click” process and forms a 1,2,3-triazole group. This mechanistic insight sheds light on the intricate chemistry occurring in the CuAAC-based reaction and contributes to a more nuanced understanding of the reaction pathway.

However, the complexity of the possible reversible and irreversible reactions poses challenges for experimental and theoretical kinetic studies over the entire membrane functionalization process. A key advantage of the additive manufacturing process is that the ink solution can be prepared to reach equilibrium before being applied. Therefore, utilizing the CuAAC reaction mechanism to introduce beneficial chemical groups (e.g., positive charges in this study) to the membrane, we divide the reaction network into two stages to study: (1) step 1 and step 2 correspond to the ink solution preparation stage, before contacting the membrane, whereas (2) step 3 illustrates the reaction mechanism isolated from transports in the membrane functionalization stage, wherein target functionality is coupled to the pore wall azide groups. Thus, an equilibrium model is suitable for steps 1 and 2 with an assumption that the volume of the ink solution is large enough to hold the concentrations constant, whereas step 3 requires a kinetic model.

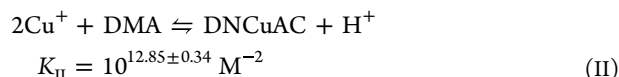
2.2. Ink Preparation Equilibria. An accurate prediction of DNCuAC concentration after the ink preparation stage (Figure 2, steps 1 and 2) is crucial for predicting the subsequent membrane functionalization progress, assuming nonzero order reactions in step 3. To ensure experimental validation of critical reaction pathways for functional membrane manufacturing, we strategically divide the reaction network into discernible steps where pH and UV–vis measurements are feasible. This work culminated in a simplified reaction network as the best ink equilibrium model in Figure 1, containing integrated reactions I and II with significant side reactions S.

Step 1



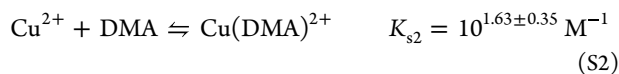
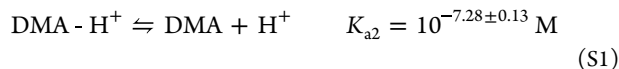
Here, label I denotes step 1 reactions in Figure 2. Reaction I1 represents the dissociation of AA, $\text{DH}_2 \rightarrow \text{DH}^-$, with its equilibrium constant K_{a1} reported by Wade.⁴⁵ Reaction I2 accounts for the conversion of Cu^{2+} to Cu^+ . The species D corresponds to the ascorbate dianion resulting from the oxidation of ascorbate anion DH^- .

Step 2



Reaction II is an integrated result of step 2 reactions, leading to the formation of our target reactant, DNCuAC. DMA denotes the alkyne-terminated reactant.

Side reactions



Additionally, denoted with S, side reaction S1 is the dissociation of 3-dimethylamino-1-propyne (DMA), $\text{DMA} - \text{H}^+ \rightarrow \text{DMA}$, with the equilibrium constant determined through pH measurements of multiple DMA solutions. Side reaction S2 is the formation of the undesired complex $\text{Cu}(\text{DMA})^{2+}$ on the dimethylamino side, which is a reaction identified beyond the main network (Supporting Information Figure S2 indicates Cu(II) –DMA complexation and Figure S3 shows UV–vis predictions with the ink equilibrium model toggling reaction S2 on and off).

Equilibrium coefficients K_{e2} , K_{II} , K_{a2} , and K_{s2} were estimated with experimental data, as described in the Materials and Methods section and Supporting Information. We report the regression results for these coefficients above (instead of in the Results and Discussion section) for brevity.

Equations 1–10 present the mathematical model describing the equilibria for reaction series I1–S2, which is derived to solve for the concentrations of all the ten species at equilibrium from any initial ink formulation of Cu(II) , AA, and DMA, denoted with subscripts 0.

Reaction equilibria

$$\frac{[\text{DH}^-][\text{H}^+]}{[\text{DH}_2]} = K_{a1} \quad (1)$$

$$\frac{[\text{Cu}^+]^2[\text{D}][\text{H}^+]}{[\text{Cu}^{2+}]^2[\text{DH}^-]} = K_{e2} \quad (2)$$

$$\frac{[\text{DNCuAC}]}{[\text{Cu}^+]^2[\text{DMA}]} = K_{\text{II}} \quad (3)$$

$$\frac{[\text{DMA}][\text{H}^+]}{[\text{DMA} - \text{H}^+]} = K_{a2} \quad (4)$$

$$\frac{[\text{Cu}(\text{DMA})^{2+}]}{[\text{Cu}^{2+}][\text{DMA}]} = K_{s2} \quad (5)$$

Conservation of Cu^{2+}

$$[\text{Cu}^{2+}]_0 = [\text{Cu}^{2+}] + [\text{Cu}^+] + 2[\text{DNCuAC}] \\ + [\text{Cu}(\text{DMA})^{2+}] \quad (6)$$

Conservation of DH_2

$$[\text{DH}_2]_0 = [\text{DH}_2] + [\text{DH}^-] + [\text{D}] \quad (7)$$

Conservation of DMA

$$[\text{DMA}]_0 = [\text{DMA}] + [\text{DMA} - \text{H}^+] + [\text{DNCuAC}] \\ + [\text{Cu}(\text{DMA})^{2+}] \quad (8)$$

Conservation of the proton

$$2[\text{DH}_2]_0 + [\text{DMA}]_0 = 2[\text{DH}_2] + [\text{DH}^-] + [\text{DMA}] + 2[\text{DMA} - \text{H}^+] + [\text{Cu}(\text{DMA})^{2+}] + [\text{H}^+] - \frac{K_w}{[\text{H}^+]} \quad (9)$$

Charge balance

$$2[\text{Cu}^{2+}]_0 = 2[\text{Cu}^{2+}] + [\text{Cu}^+] + [\text{DNCuAC}] + 2[\text{Cu}(\text{DMA})^{2+}] + [\text{DMA} - \text{H}^+] + [\text{H}^+] - [\text{DH}^-] - \frac{K_w}{[\text{H}^+]} \quad (10)$$

As a matter of computational accuracy and efficiency, a reformulated model (i.e., log-transformed) is provided in the [Supporting Information](#). This model serves as an effective tool for predicting the concentration of DNCuAC.

2.3. UV–Vis Spectrophotometry for Multicomponent Systems. Due to the importance of copper in the reaction network, it is essential to know the concentration of copper-containing species in order to track the reaction progress. The concentration of an analyte that absorbs light, e.g., Cu(II), can be obtained through Beer–Lambert law, i.e., the absorbance measurement is linearly proportional to the molar concentration of the sample at a specific wavelength

$$A_\lambda = \epsilon_\lambda b C \quad (11)$$

where A_λ represents the UV–vis absorbance at wavelength λ , ϵ_λ denotes the molar absorptivity or molar extinction coefficient, b is the length of the light path through the sample, and C represents the molar concentration of the component being determined. In a multicomponent mixture where molecules do not interact with one another, the total absorbance $A_{\lambda,t}$ of n components at a given wavelength λ is the sum of individual absorbances $\sum_i A_{\lambda,i}$ attributed to each chromogenic component i in the mixture

$$A_{\lambda,t} = \sum_{i=1}^n A_{\lambda,i} = \mathbf{k}^T \mathbf{C} \quad (12)$$

where $\mathbf{k} = [\epsilon_{\lambda,1}b, \epsilon_{\lambda,2}b, \dots, \epsilon_{\lambda,n}b]^T$ and $\mathbf{C} = [C_1, C_2, \dots, C_n]^T$. The proportional constants $\mathbf{k}$ can be estimated using multiple linear regression with single standards or mixtures of components. Concentrations of unknown mixtures, $\mathbf{C}$, can be determined by solving the linear system with several absorbance readings.

2.4. Instantaneous Reaction Model for the Membrane Functionalization Process. In the membrane functionalization stage, immobilized azide moieties lining the pore walls of the membrane are functionalized by placing a reactive ink solution on top of the membrane surface and controlling the time that the reactive molecules diffuse into the polymer matrix. The reaction between immobilized azide moieties and the diffusing ink components is described by reaction step 3 ([Figure 2](#)). Because the azide moieties are immobilized, diffusion of the active click reactant, e.g., DNCuAC, into the membrane is the dominant transport mechanism. The interplay between the reaction rate (with a pseudo first-order approximation) and diffusion rate in the functionalization process can be characterized by the Damköhler number D_a , which is the ratio of rates of these

two phenomena. The numerator of D_a is the product of the reaction rate constant k_R and the concentration of the parent azide. The denominator is the diffusion coefficient of the reactant in the membrane D divided by the squared characteristic length associated with diffusion, l , i.e., the membrane thickness⁴⁶

$$D_a = \frac{k_R C_{A0} l^2}{D} \quad (13)$$

When $D_a \gg 1$, the reaction occurs much faster than the species diffuses through the membrane. As such, the reaction is said to be instantaneous. These high Damköhler numbers represent the reactions that are desired to systematically introduce multiple moieties throughout the membrane.²¹ This assumption implies that the functionalized depth is directly constrained by the diffusion depth and can be controlled by adjusting the depositing time of the ink solution.

In the limit of $D_a \gg 1$, Hoffman et al.²¹ proposed

$$\frac{d(A_m C_A (L - L'))}{dt} = - \left[\frac{D}{L'} (C_D - 0) \right] A_m \quad (14)$$

Here, the rate at which the azide moieties are consumed (i.e., the left-hand side of [eq 14](#)) is assumed to be equal to the rate at which the reactant species, DNCuAC, diffuses through the membrane as a penetration front (i.e., right-hand side of [eq 14](#)). A_m is the membrane area exposed to the reactive solution, L is the membrane thickness, L' is the penetration depth that defines the ink–azide interface location, t is the ink penetrating time, C_A denotes the concentration of azide, and C_D is the concentration of active reactant DNCuAC in the depositing ink solution. Given that the volume of the deposition ink solution significantly exceeds the membrane volume, C_D is assumed to remain constant in the bulk solution. Under the assumption of instantaneous reaction, the azide concentration profile can be simplified into a stepwise function. Specifically, in regions reached by the ink solution (i.e., at lengths smaller than the penetration front), all azide moieties have been consumed, while any unreached region maintains an azide concentration equal to its initial value C_{A0}

$$C_A(t, z) = 0 \quad z < L' \\ C_A(t, z) = C_{A0} \quad z \geq L' \quad (15)$$

This simplification enables easy analytical integration of the azide functionalization, along with an initial condition $L'(t = 0) = 0$; L' can be calculated using the stepwise concentration profile of C_A at any penetrating time

$$L'(t) = \sqrt{\frac{2DC_D t}{C_{A0}}} \quad (16)$$

The instantaneous reaction model [eqs 14–16](#) serves as the first candidate model for step 3 described in [Figure 1](#).

2.5. Reaction-Diffusion PDE Model for the General Functionalization Process. We propose a more general functionalization model by relaxing the instantaneous reaction assumption in [Section 2.4](#). This more general approach provides an alternative candidate model for step 3. It captures both the diffusion dynamics and reaction kinetics of the DNCuAC, denoted by its concentration C_D

$$\frac{\partial C_D}{\partial t} = D \frac{\partial^2 C_D}{\partial z^2} - k_a C_A C_D \quad (17)$$

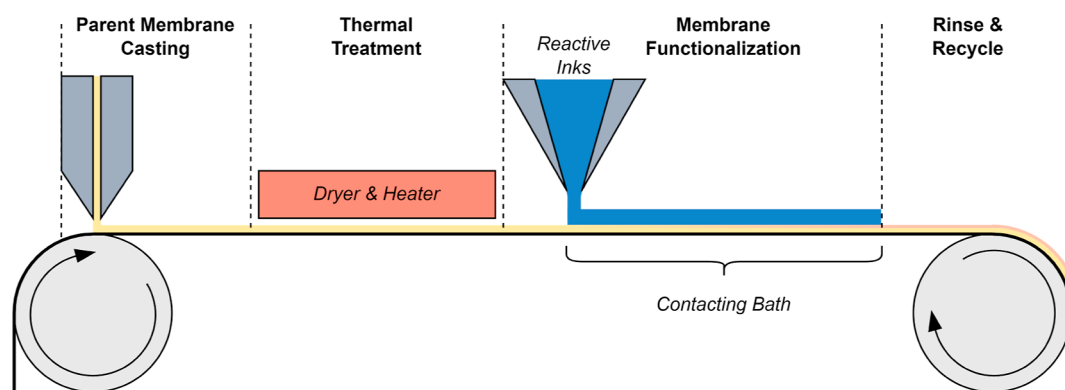


Figure 3. Schematic illustration of the additive manufacturing process of functional membranes.

where t represents the time elapsed since the ink solution contacts the membrane surface, z signifies the diffusion depth, D is the diffusion coefficient of DNCuAC in the membrane phase, k_a is the overall reaction rate constant for step 3 in the CuAAC network in Figure 2, and C_A denotes the azide concentration on the membrane. At the onset, the absence of DNCuAC in the membrane gives the following initial condition

$$C_D(0, z) = 0, \quad z \neq 0 \quad (18)$$

Again, we assume that the ink droplet is sufficiently large (e.g., ~ 100 -fold volume difference) to maintain a constant DNCuAC concentration C_{D0} at the top boundary of the membrane

$$C_D(t, 0) = C_{D0} \quad (19)$$

At the lower boundary of the membrane (i.e., $z = L$), no concentration gradient exists due to the impenetrable contact between the membrane and the supporting substrate

$$\left. \frac{\partial C_D}{\partial z} \right|_{z=L} = 0 \quad (20)$$

The azide anchored on the membrane pore wall with its concentration profile is only subject to the step 3 reaction, thus

$$\frac{\partial C_A}{\partial t} = -k_a C_A C_D \quad (21)$$

The initial condition is given by the uniform azide concentration C_{A0} of the unfunctionalized parent membrane

$$C_A(0, z) = C_{A0} \quad (22)$$

The PDE model eqs 17–22 describe the concentration profiles of mobile DNCuAC and immobile azide which are coupled in the reaction term. Supporting Information eqs S11–S18 describe the nondimensionalized version of the PDE model we solved numerically.

2.6. FTIR Measuring Azide Conversion. At a given wavenumber, the intensity of the absorbance signal A from FTIR spectroscopy can effectively capture the integral of the azide concentration profile, and it is governed by the following relationship⁴⁷

$$A(t, L) = \int_0^L \epsilon C_A e^{-2\gamma z} dz \quad (23)$$

where the emissivity of the sample, ϵ , and the inverse of the depth that the evanescent wave of the FTIR laser penetrates

into the sample, γ , are two constants. The conversion of azide, denoted as η , is defined as

$$\eta = 1 - \frac{A}{A_{\text{parent}}} \quad (24)$$

where A is the measured FTIR intensity after the reaction has progressed for time t and A_{parent} is the measured FTIR intensity of the parent membrane ($t = 0$). Analytically integrating eq 23 under the assumption of an instantaneous reaction ($D_a \gg 1$) establishes a relationship between the absorbance signal intensity and the reactive penetration depth, L'

$$A(t, L) = \frac{\epsilon C_{A0}}{2\gamma} [e^{-2\gamma L'(t)} - e^{-2\gamma L}] \quad (25)$$

By substitution of eqs 16 and 25 into eq 24, the azide conversion can be expressed as a function of time t for the membrane functionalization process with the assumption of instantaneous reaction

$$\eta_{D_a \gg 1} = \frac{1 - e^{-2\gamma L'(t)}}{1 - e^{-2\gamma L}} = \frac{1 - e^{-2\gamma \sqrt{\frac{2DC_{D0}t}{C_{A0}}}}}{1 - e^{-2\gamma L}} \quad (26)$$

Moreover, when the membrane is thick enough, i.e., $L > 5 \mu\text{m}$, eq 26 can be rearranged into

$$-\ln(1 - \eta_{D_a \gg 1}) \sqrt{\frac{C_{A0}}{C_D}} = \sqrt{8D} \gamma \sqrt{t} \quad (27)$$

which implies a linear relationship between the left-hand side and the square root of time.

In contrast, efficient numerical integration strategies are essential to evaluate FTIR absorbance and azide conversion for reaction-diffusion PDEs in the general functionalization process.

2.7. Ink Optimization Problem. Optimization problem eqs 28a–28c determines the best reactive ink formulation $\mathbf{x} = ([\text{Cu}^{2+}]_0, [\text{DH}_2]_0, [\text{DMA}]_0)^T$ that minimizes the relative economic or environmental cost of the ink materials, $\mathbf{w}^T \mathbf{x}$. For example, we use weights $\mathbf{w}^T = [1, 0, 0]$ to prioritize the reduction of copper usage in the membrane functionalization process.

$$\min_{\mathbf{x}} \mathbf{w}^T \mathbf{x} \quad (28a)$$

$$\text{s. t. } C_{D0} = f_{1,2}(\mathbf{x}) \quad \text{step 1 and 2} \quad (28b)$$

$$\eta(t_{\text{end}}) = f_3(C_{\text{D}0}, C_{\text{A}0}, L, t_{\text{end}}) \geq \eta^* \quad \text{step 3} \quad (28c)$$

Constraint eq 28c ensures that the functionalization conversion requirement η^* is achieved or exceeded at designed processing time t_{end} that the reactive inks are rinsed off to stop the functionalization reaction. Here, $f_3(\cdot, \cdot, \cdot, \cdot)$ refers to the step 3 (see Figure 2) membrane functionalization model, either eqs 14–16 (fast reaction assumption) or eqs 17–22 (rigorous PDEs), which predicts the conversion η as a function of DNCuAC concentration $C_{\text{D}0}$, parent membrane azide concentration $C_{\text{A}0}$, membrane thickness L , and t_{end} . Constraint eq 28b uses the steps 1 and 2 ink preparation model $f_{1,2}(\cdot)$, eqs 1–10, to predict the DNCuAC concentration at equilibrium $C_{\text{D}0} = [\text{DNCuAC}]_{\text{eq}}$ from any initial ink solution $\mathbf{x}$.

We now consider a roll-to-roll membrane manufacturing process, illustrated in Figure 3, to demonstrate the flexibility of the ink optimization framework. The processing speed is primarily determined by the post-film-formation thermal treatment and typically ranges from 0.1 to 200 m·min^{−1}.⁴⁸ Integrating functionalization steps following the parent membrane fabrication allows for the continuous large-scale manufacturing of multifunctional membranes.¹⁶ The functionalization time t_{end} will be converted to bath length L_{bath} for ink solution–membrane contact in the process design. For example, continuous membrane manufacturing at 5 m·min^{−1}, using a 1 min functionalization process, requires 5 m long bathing. From Figure 2, CuAAC mechanisms provide conservation of copper with Cu(I) released back to the ink solution in step 3. Assuming that copper can be fully recycled, continuous operation with a known rolling speed will require a total copper investment $U_{\text{Cu}^{2+}}$ proportional to initial copper concentration and the processing time t_{end}

$$U_{\text{Cu}^{2+}} \propto [\text{Cu}^{2+}]_0 \cdot L_{\text{bath}} \propto [\text{Cu}^{2+}]_0 \cdot t_{\text{end}} \quad (29)$$

Minimizing the total copper investment corresponds to replacing objective eq 28a with

$$\begin{aligned} \arg \min_{\mathbf{x}, t_{\text{end}}} U_{\text{Cu}^{2+}} &= \arg \min_{\mathbf{x}, t_{\text{end}}} [\text{Cu}^{2+}]_0 \cdot t_{\text{end}} \\ &= \arg \min_{\mathbf{x}, t_{\text{end}}} \mathbf{w}^T \mathbf{x} \cdot t_{\text{end}} \end{aligned} \quad (30)$$

while constraints eqs 28b and 28c stay the same. The ink optimization problems accommodating other considerations are extendable to other multifunctional membrane manufacturing systems, e.g., surface-patterned membranes.¹⁹ Weight selection can be informed by life cycle assessment (LCA) of a specific process design.

3. MATERIALS AND METHODS

3.1. Materials and Equipment. All chemicals, unless otherwise noted, were obtained from Sigma-Aldrich. All aqueous solutions used deionized water (DI water) obtained from a Millipore water purification system (Milli-Q Advantage A10, Milli Q, MA). Copper(II) sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), AA, and DMA were used in the ink preparation for membrane functionalization.

3.2. Cu(II)-AA-DMA pH Measurements. Kinetic pH data after mixing the alkyne, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, and AA were obtained using a 115 Accumet Portable pH probe (Fischer Scientific), with constant measurement error ± 0.01 . The concentrations of AA and copper sulfate were held constant at 150 mM and 50 mM, respectively. The concentration of the alkyne varied between 25, 50, and 100 mM. Specifically, AA was initially mixed with the alkyne, and the pH was recorded. Nitrogen gas, continuously bubbled into the solution, prevented oxygen from dissolving into the solution and kept the solution well-mixed. After equilibrium was reached, copper sulfate was

added, diluting the mixture to the desired concentration. The pH of the solution was recorded at 15 s intervals until the new equilibrium had been reached. Between experiments, the pH probe was rinsed with DI water and stored in a pH 7 buffer solution. All experiments were conducted in duplicate.

3.3. Cu(II)-AA-DMA UV–Vis Measurements. UV–vis data for the mixing of DMA, AA, and copper sulfate pentahydrate was obtained using the “Kinetics” mode of a Cary 60 UV–vis spectrometer (Agilent Technologies, software version: 5.0.0.999). Measurements were taken in disposable 1 mL cuvettes with reading error ± 0.0007 . The instrument was zeroed using 1 mL of DI water. The concentration of AA and copper sulfate was held constant at 150 mM and 50 mM, respectively. The concentration of the alkyne varied between 25, 50, and 100 mM. All experiments were performed in duplicate.

Before the experiments were run, the “Scan” feature of the UV–vis spectrometer was used to identify the wavelength of maximum absorbance for copper sulfate pentahydrate (i.e., 814 nm). To obtain the kinetic data, AA and DMA were mixed in a 1 mL cuvette. At least 10 min were allowed for the solution to come to equilibrium. Subsequently, copper sulfate pentahydrate was added, and the UV–vis measurements were started immediately after mixing. Measurements were obtained continuously until the solution came to equilibrium (~ 10 min).

3.4. Membrane Fabrication. The P(TFEMA-OEGMA-AHPMA) copolymer was synthesized according to previous methods.⁴⁹ An 18 wt % casting solution was made by dissolving the copolymer in trifluoroethanol (TFE). The resulting solution was passed through 1 μm glass filters and degassed by stirring at 60 rpm for at least 24 h. Subsequently, ~ 1 mL of solution was placed on a commercial poly(vinylidene fluoride) ultrafiltration membrane support (Nanostone Water, Inc., PV400 ultrafiltration membrane). The membrane was cast over the support using a doctor blade set at a gate height of 70 μm . The TFE solvent was allowed to evaporate for 5 min ± 10 s, and then, the membrane was submerged into a DI water bath, precipitating the copolymer and fixing its nanostructure in place. Membranes were subsequently cut into 1 in. diameter samples and stored in DI water until further use. The final concentration of the azide moieties was 850 ± 15 mM.

3.5. Membrane Functionalization and FTIR Measurements. Reactive ink solutions were formed by using DMA, copper sulfate pentahydrate, and AA. The $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ concentration was held at 10, 50, and 100 mM. Three ratios of DMA to $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 1:2, 1:1, and 2:1, were used to prepare samples. The concentration of AA was held at 3 $\times$ excess of the concentration of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ for all solutions. Membranes were cut into areas of roughly 0.7 cm² and placed into Petri dishes. The reactive solution mixing process was similar to the process used for UV–vis measurements, in which case after adding DMA to the reactive solution and waiting ~ 10 min to reach equilibrium, it was deposited onto the membrane surface. 100 μL of reactive solution was deposited on membrane samples and was allowed to react for a set amount of time which ranged from 20 to 120 s. After the set time, the membrane and reactive solution were submerged in DI water to stop the reaction. Membranes were dried by placing them in a vacuum oven at room temperature overnight.

A Jasco 6300 FTIR spectrometer was used for FTIR analysis. Each membrane sample was scanned 25 times from 650 to 4000 cm^{−1}. Conversion of the azide moieties was monitored by using the decrease in intensity of the azide peak at 2100 cm^{−1}. All spectra were normalized using the carbonyl peak at 1725 cm^{−1}, which is present within the backbone of the parent copolymer and remains unaffected throughout the functionalization process. For the experimental conditions in this paper, FTIR penetration depth was estimated and assumed to be a constant for all calculations.²¹ $\gamma = 0.98 \mu\text{m}^{-1}$.

3.6. Data Science Framework. The ink optimization problems eqs 28a–28c and 30 were solved by integrating a range of experimental and data science techniques, which are described below.

3.6.1. Model Parameter Estimation. We start by estimating the parameters in models $f_{1,2}$ and f_3 . As a generalized notation, let $\hat{y}_{r,i} = f(\mathbf{x}_i, \mathbf{u}_i, \boldsymbol{\theta})$ represent the model prediction with varied

Table 1. Parameter Estimation and Corresponding Experiments

experimental system	y	x	u	θ
Cu(II)-AA (Supporting Information)	pH	$[\text{Cu}^{2+}]_0$	$[\text{DH}_2]_0$	K_{e2}
Cu(II)-AA-DMA	pH, UV-vis	$[\text{DMA}]_0$	$[\text{Cu}^{2+}]_0, [\text{DH}_2]_0$	K_{II}
DMA	pH	$[\text{DMA}]_0$		K_{a2}
Cu(II)-DMA complex (Supporting Information)	UV-vis	$[\text{Cu}^{2+}]_0, [\text{DMA}]_0$		K_{s2}
FTIR-informed azide	azide conversion	$[\text{Cu}^{2+}]_0, [\text{DH}_2]_0, [\text{DMA}]_0$	C_{A0}, L	D, k_{a}

experimental conditions x_r , constant/controlled experimental conditions u_r , and model parameter values θ . Here, r is the index for the type of response and i is the index for the experiment number. Furthermore, we assume that the measured observation $y_{r,i}$ is the model prediction $\hat{y}_{r,i}$ with an additive observation error $\epsilon_{r,i}$

$$y_{r,i} = \frac{f(x_{r,i}, u_{r,i}, \theta)}{\hat{y}_{r,i}} + \epsilon_{r,i}, \quad \epsilon_{r,i} \sim \mathcal{N}(0, \sigma_r^2) \quad (31)$$

For this study, we assume that the observation errors $\epsilon_{r,i}$ follow independent Gaussian distributions with mean zero and variance σ_r^2 . For convenience, we stack all of the measurements and experimental conditions into vectors, e.g., $y = [y_{r,i} : \forall r \in \{1, \dots, R\}, i \in \{1, \dots, n_r\}]^T$. We use weighted least-squares (WLS) nonlinear regression to compute the estimate $\hat{\theta}$ from data x, y , and u

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^p} \sum_{r=1}^R \frac{1}{n_r} \sum_{i=1}^{n_r} w_{r,i} (y_{r,i} - \hat{y}_{r,i})^2 \quad (32)$$

Here, n_r is the number of observations for response r and the total number of observations is $N = \sum_{r=1}^R n_r$. The squared residuals are weighted by the inverse of the variance of measurement errors, $w_{r,i} = 1/\sigma_r^2$. Table 1 summarizes the five types of experiments analyzed in this work. The equilibrium parameters are log-transformed (see Supporting Information), which both improves the numeric performance and enforces the parameters to be positive. D and k_{a} are explicitly bounded in eq 32 to be non-negative.

Minimizing the WLS objective is equivalent to maximizing the probability of observing the data y given model parameters θ (under our assumed error model).⁵⁰ This probability is known as the likelihood function, $\mathcal{L}(\theta|y)$, which is mathematically defined in the Supporting Information.

3.6.2. Uncertainty Quantification and Estimability Analysis. Next, we consider how the measurement errors $\epsilon_{r,i}$ propagate into uncertainty in parameter estimate $\hat{\theta}$. This uncertainty is quantified by the parameter covariance matrix $V_{\hat{\theta}}$.⁵¹

$$V_{\hat{\theta}} \approx (Q^T \Sigma_y^{-1} Q)^{-1} \quad (33)$$

Here, Q is the matrix of local sensitivities of the model predictions, i.e., partial derivatives of the model predictions with respect to each parameter⁵² evaluated at the parameter estimate $\hat{\theta}$. Likewise, Σ_y is the covariance matrix of measurement errors, which is diagonal under the independence assumption.

Mean model prediction uncertainties can be approximated through first-order propagation⁵³

$$\Sigma_{\hat{y}} \approx \bar{Q} V_{\hat{\theta}} \bar{Q}^T \quad (34)$$

Here $\bar{Q}$ is the sensitivity matrix evaluated at the prediction conditions. Similarly, elliptical confidence regions of parameters can be approximated with⁵¹

$$(\theta - \hat{\theta})^T V_{\hat{\theta}}^{-1} (\theta - \hat{\theta}) \leq p F_{(1-\alpha, p, N-p)} \quad (35)$$

where $F_{(1-\alpha, p, N-p)}$ is the 100(1 - α)% point of the F distribution.

For especially nonlinear models, it is significantly more accurate to construct the confidence regions via the likelihood ratio test^{54,55}

$$2(\ln(\mathcal{L}(\hat{\theta}|x)) - \ln(\mathcal{L}(\theta|x))) \leq \eta_{bc} \chi_{1-\alpha, p}^2 \quad (36)$$

where $\chi_{1-\alpha, p}^2$ is the 100(1 - α)% point of the χ^2 distribution and η_{bc} is the Bartlett correction factor, which we set as one in this work.

The Fisher information matrix (FIM), M , measures the information content of measurements y at experimental conditions x and u for inferring parameters θ in mathematical model f . The FIM is approximately the inverse of the covariance matrix³⁶

$$M \approx V_{\hat{\theta}}^{-1} \approx Q^T \Sigma_y^{-1} Q \quad (37)$$

A singular FIM indicates that the model is locally not practically identifiable.⁵⁶ Decomposing M into eigenvalues Λ and eigenvectors S can reveal which linear combinations of parameters cannot be reliably estimated

$$M = S^T \Lambda S \quad (38)$$

3.6.3. Model Selection. For any alternate reaction pathways, we can posit a new candidate mathematical model. For example, toggling the presence or absence of side reaction S2 and another possible byproduct $\text{Cu}(\text{DMA})_2^{2+}$ results in four model variations for $f_{1,2}$. Adhering to the principle of model parsimony, we employed the Akaike information criterion (AIC)⁵⁷ in model selection to discriminate the most plausible reaction pathway. The AIC integrates both model fits, i.e., likelihood $\mathcal{L}(\hat{\theta}|y)$, and a penalty through a measure of model complexity k to safeguard against overfitting

$$\text{AIC} = -2\ln(\mathcal{L}(\hat{\theta}|y)) + 2k \quad (39)$$

For the model calibrated through eq 32 with R types of measurements and p parameters, assuming each type of measurement has an independent constant error (see full derivation in the Supporting Information), we have

$$\begin{aligned} \text{AIC} = & (\ln(2\pi) + 1) \sum_{r=1}^R n_r + \sum_{r=1}^R \left(n_r \ln \left(\frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) \right) \\ & + 2k, \quad k = p + R \end{aligned} \quad (40)$$

In AIC ranking, only differences in AIC are meaningful for model comparisons; therefore, dropping the constant term leaves a convenient ΔAIC formula

$$\Delta\text{AIC} = \sum_{r=1}^R \left(n_r \ln \left(\frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) \right) + 2(p + R) \quad (41)$$

Considering the small observation size with respect to the number of system parameters, i.e., $N/k < 40$, a correction was introduced to the AIC^{58,59}

$$\Delta\text{AIC}_c = \Delta\text{AIC} + \frac{2k(k+1)}{N-k-1} \quad (42)$$

The model with the lowest ΔAIC value is generally considered as an optimal model form providing both good predictions and parsimonious representations of the system.

3.6.4. Design of Experiments. Design of experiments principles guide the suggestion of experimental conditions and data collection. To regress models $f_{1,2}$ and f_3 in eqs 28b and 28c, we followed the 3^k

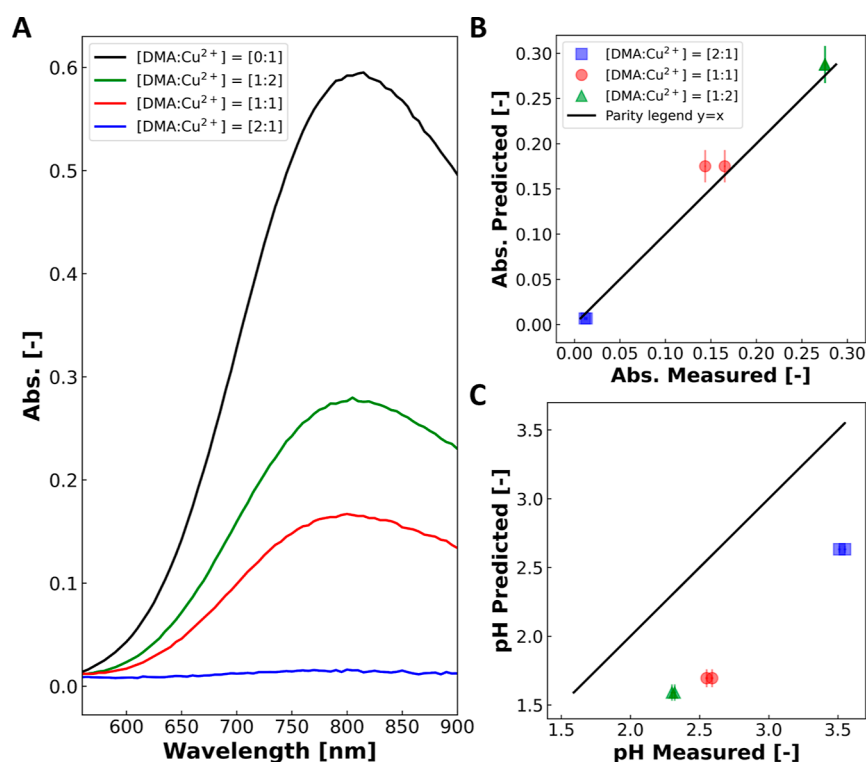


Figure 4. Comparing predictions from the ink preparation equilibrium model with experimental measurements. UV–vis absorbance measurements (A) are compared with predictions (B) calculated using eq 12 for the calibrated components Cu²⁺ and Cu(DMA)²⁺. pH predictions (C) are calculated from the concentration predictions of H⁺ in the model. All ink solutions, measured in duplicates, contain a [AA/Cu²⁺] = [3:1] ratio, and the following [DMA/Cu²⁺] ratios of [2:1], [1:1], and [1:2] with Cu²⁺ concentration at 50 mM. Horizontal error bars $\epsilon_{UV-vis} = \pm 0.0007$ and $\epsilon_{pH} = \pm 0.01$ are not noticeable on the plots.

factorial design⁶⁰ where k in this context is the number of factors, i.e., dimensions of $\mathbf{x}$ in Table 1. This experimental design involves varying the levels of each factor across three levels, allowing for the exploration of the main effects as well as interactions between factors. For example, the Cu(II)-AA-DMA mixture used for pH and UV–vis measurements was prepared at 3 levels of Cu(II)/DMA ratios (Sections 3.2 and 3.3), and FTIR measurements were taken with one more factor, i.e., 3-level Cu(II) usage (Section 3.5). In the Supporting Information, we demonstrated that a 3^k factorial design was sufficient to identify the Cu(II)–DMA complexation. By systematically varying the ink components according to this factorial design, we were able to comprehensively model the influence of different component ratios on the ink equilibria and membrane functionalization process.

Through the careful data analysis framework, we were able to gain valuable insights into the ink formulation driving the membrane functionalization process and optimize the design of the multifunctional membrane manufacturing processes.

3.7. Computational Environments. All modeling and simulations were implemented with the open-source Pyomo package (version 6.6.2)⁴¹ in Python (version 3.11.6). The PDE version of the step 3 model f_3 was numerically integrated via a backward finite difference method using Pyomo.DAE.⁴² Parallel computing on HPC clusters using the Multiple Processing Interface (MPI) through the mpi4py package (version 3.1.5)⁶¹ enabled sensitivity studies to efficiently estimate the PDE model parameters and quantify their uncertainty. For illustration, a typical simulation time evaluating azide conversion over a single fixed ink formulation ranges from 10 to 200 s on a laptop equipped with Intel Core i7-1065G7 CPU at 1.30 GHz, with 32 GB of RAM, and the Windows 10 64-bit operation system. With efficient scheduling in the RedHat Linux 8.9 operation system on the high-performance computing resources at the Notre Dame Center for Research Computing (ND-CRC), the model calibration and exploration of ink design space were completed within minutes on multiple cores (e.g., Dual 32-core AMD EPYC 7543 CPU at 2.80

GHz with 256 GB RAM). These open-source packages not only enable transferable and reproducible model implementation but also provide access to advanced data analysis tools (e.g., parameter estimation⁶² and design of experiments³⁴) and state-of-the-art optimization algorithms (e.g., Ipopt version 3.13.2⁶³ with HSL MA27 linear solver⁶⁴ distributed via the IDAES-PSE package⁶⁵).

4. RESULTS AND DISCUSSION

We now show how the data science workflow enables the modeling and optimization of the ink for the membrane functionalization process.

4.1. Ink Equilibrium Model Predicts Concentration of the Catalytic-Active Component in CuAAC. Calibrated equilibrium constants and associated uncertainties are listed in eqs I1 to S2 in Section 2.2. In this section, we focus on comparing the predictions from this best model to pH and UV–vis measurement data from the Cu(II)-AA-DMA mixture (see Sections 3.2 and 3.3), which were used to calibrate the step 2 reaction. Refer to the Supporting Information for more information about calibrating the step 1 and side reaction equilibrium constants. Further efforts to develop kinetic models for steps 1 and 2 were unsuccessful because the reaction proceeded too quickly to obtain sufficient data, as seen both in this study and in the existing literature.

Figure 4A shows representative data sets of the UV–vis absorbance curves. A solution of 50 mM Cu²⁺ (black line) provides a characteristic absorbance peak at ~815 nm. Keeping the Cu²⁺ concentration constant and increasing the DMA concentration lead to a decrease in the absorbance peak, signifying that copper is consumed by the DMA to form complexes. The parity plot in Figure 4B shows that the

absorbance measurements are in close alignment with those predicted by the fully calibrated ink preparation equilibrium model derived from eqs 1–10. Figure 4C, on the other hand, shows that the model predictions for pH from the same formulations (as described in Section 3.1) are less accurate. The vertical model prediction error bars were calculated by propagating measurement uncertainties in the K_{II} estimate via eq 34.

Figure 4B shows that the model accurately predicts the combined absorbance of Cu^{2+} and $\text{Cu}(\text{DMA})^{2+}$ for various ink formulations, i.e., $[\text{DMA}/\text{Cu}^{2+}]$ ratios. Using these predictions and assuming that copper is conserved, we are able to then predict the concentration of DNCuAC. Figure 4C, on the other hand, illustrates the limitation of the ink preparation equilibrium model in predicting the pH of the system as it overestimates the concentration of hydrogen ions by approximately an order of magnitude. We suspect that the inaccurate pH predictions are because the model assumes a full conversion and no significant formulation of intermediates such as MCuPIC in the main reaction network. However, these intermediates, if present, could hinder the generation of hydrogen ions. While more direct measurement to DNCuAC formation or optimal designed experiments could provide better insights into the detailed ink reaction pathways, it is reassuring to note that the ink preparation equilibrium model performs well in the integrated prediction of the functionalization process within the ink optimization work.

To help determine the reaction pathways and possible intermediates in model $f_{1,2}$, we employed practical identifiability (i.e., estimability) analyses and model selection for step 1 and side reaction S2, as provided in Supporting Information. In Supporting Information Table S1, the eigen decomposition of the FIM shows that the equilibrium of a $\text{Cu}(\text{II})$ –AA complex formation (reaction i2 in Supporting Information) is not practically identifiable. Consequently, we simplified step 1 to two reactions I1 and I2 integrated from the main reaction network, which passed the identifiability analyses and demonstrated excellent capability in predicting the pH of a wide range of $\text{Cu}(\text{II})$ –AA mixtures, as shown in Supporting Information Figure S1. Table 2 presents the AIC_c rankings for

Table 2. AIC_c Ranking for $\text{Cu}(\text{II})$ –Alkyne Complex Forms

rank	complex form	WLS objective	model complexity k	ΔAIC	ΔAIC_c
1	$\text{Cu}(\text{DMA})^{2+}$	10.79	4	29.41	39.41
2	$\text{Cu}(\text{DMA})_2^{2+}$	12.73	4	30.90	40.90
3	no complex	98.83	2	45.34	47.34
4	$\text{Cu}(\text{DMA})^{2+}$ and $\text{Cu}(\text{DMA})_2^{2+}$	10.05	6	32.77	74.77

candidate models with different complex forms using the same set of $\text{Cu}(\text{II})$ –DMA UV–vis data, indicating that $\text{Cu}(\text{DMA})^{2+}$ is the most probable complex from a statistical point of view based on the available data. The mathematical model assuming the presence of both complexes $\text{Cu}(\text{DMA})^{2+}$ and $\text{Cu}(\text{DMA})_2^{2+}$ in $\text{Cu}(\text{II})$ –DMA solution is provided in the Supporting Information. Supporting Information Figure S3 shows that incorporating this complex with side reaction S2, the equilibrium model is capable of correcting biases in UV–vis predictions by propagating the influence of the additional chromogenic complex $\text{Cu}(\text{DMA})^{2+}$ via eq 12.

4.2. Calibrating the Reaction PDE Model with FTIR-Informed Conversion Measurements. Parameters D and

k_a in eqs 17 and 21 characterize the functionalization process and can be regressed using FTIR data. Figure 5 presents the

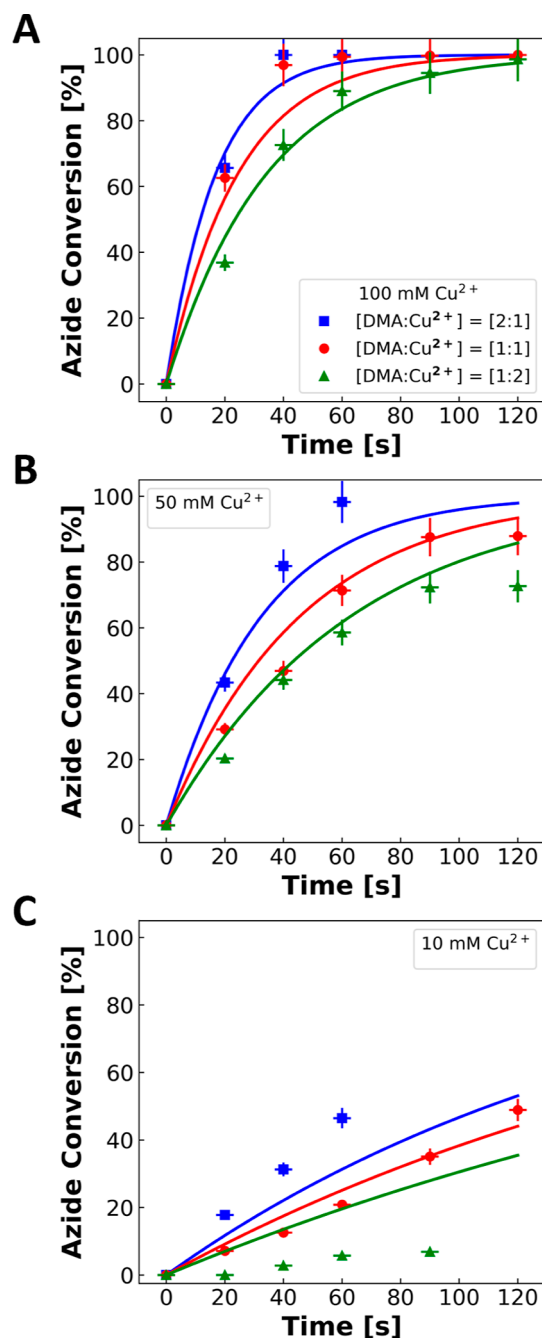


Figure 5. Time-series azide conversion data points compared to computational predictions in solid lines for the CuAAC functionalization processes using 9 different ink formulations. All solutions contained a $[\text{AA}/\text{Cu}^{2+}] = [3:1]$ ratio and the following $[\text{DMA}/\text{Cu}^{2+}]$ ratios of [2:1], [1:1], and [1:2] with Cu^{2+} concentrations at (A) 100, (B) 50, and (C) 10 mM. The model prediction uncertainty bands which are approximately $\pm 1.5\%$ via eq 34 are not noticeable.

nine sets of azide conversion data points quantified by FTIR intensities using eq 24. Each data point in Figure 5 is accompanied by cross-shaped error bars: the vertical error bars account for deviations in readings calculated by propagating the error associated with the intensities of the reacted and unreacted samples; the horizontal error bars accommodate a

± 3 s variation in preparing sample (i.e., rinsing to terminate the reaction) for FTIR processing. Since the functionalization reaction can be very fast, the uncertainty resulting from the measurement time of FTIR becomes significant. Parameter estimation only includes the error information from the FTIR measurements and yields the best-fit result in Table 3, denoted

Table 3. Parameter Results of the Reaction-Diffusion PDE Model

D [$\mu\text{m}^2\cdot\text{s}^{-1}$]	k_a [$\text{M}^{-1}\cdot\text{s}^{-1}$]	covariance $V_{\hat{\theta}}$
6.3	1.69	$\begin{bmatrix} 0.41 & -0.015 \\ -0.015 & 5.8 \times 10^{-4} \end{bmatrix}$

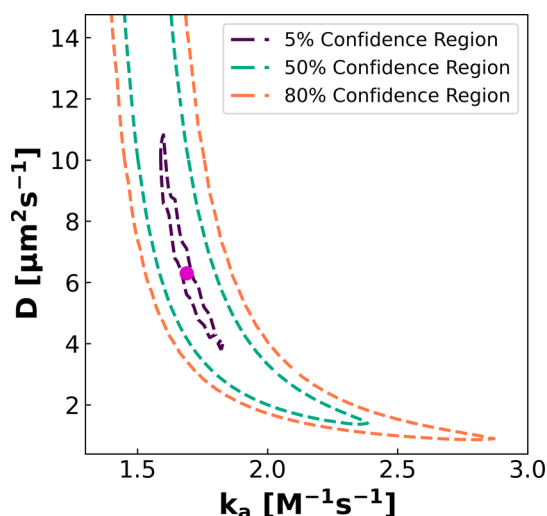


Figure 6. Likelihood ratio confidence regions for diffusion coefficient D and kinetic parameter k_a in the reaction-diffusion PDE model. Confidence levels are calculated by eq 36.

as magenta points in Figure 6 and Supporting Information Figure S4, where $D = 6.3 \pm 0.7 \mu\text{m}^2\cdot\text{s}^{-1}$ and $k_a = 1.69 \pm 0.03 \text{ M}^{-1}\cdot\text{s}^{-1}$. The Damköhler number was calculated through eq 13 for this functionalization process with Cu(II)-AA-DMA ink solutions, $D_a = 51.27 \pm 5.77$, representing a relatively faster reaction over diffusion. Here, the membranes used for FTIR analysis are of thickness $L = 15 \mu\text{m}$.

Figure 6 shows the confidence regions calculated from the likelihood ratio test, eq 36, whose shapes are derived from the WLS objective contours in Supporting Information Figure S4. For example, all parameter combinations in the purple dashed region in Supporting Information Figure S4 provide a good quality of fit with up to 1% difference in the objective, corresponding to a 20% confidence level in Figure 6. It is noteworthy that the diffusion coefficient D is only well-bounded at low confidence levels, $\leq \sim 5\%$. The left-slanted narrow shapes of these confidence regions indicate low confidence in the D value estimated from the conversion data informed by FTIR and its highly negative correlation with k_a around the best fit, which is consistent with the covariance presented in Table 3.

For comparison, linear regression was performed for the instantaneous reaction model eq 27 using the high conversion experimental data, which resulted in the estimate of $D = 0.9$

$\mu\text{m}^2\cdot\text{s}^{-1}$. This result is comparable with the diffusion coefficients for propiolic acid and propargyl chloride reported by Hoffman et al.²¹ We will instead focus on comparing the diffusion coefficients across the PDE ($6.3 \mu\text{m}^2\cdot\text{s}^{-1}$) and instantaneous reaction ($0.9 \mu\text{m}^2\cdot\text{s}^{-1}$) models. The effective diffusion coefficient in the instantaneous reaction model is smaller because it must also account for any real reaction rate limitations. This smaller D (instantaneous reaction model) agrees with the long right bottom tail of the $\geq 80\%$ confidence region of D and k_a in Figure 6 (PDE model). The unbounded D on the other side of the confidence region represents systems where the azide conversion is dominated by the reaction rate. In this region, the model predictions are highly sensitive to k_a and insensitive to D . This also explains why our estimate of D for the instantaneous reaction model has a large uncertainty. This finding demonstrates the great potential of using FTIR as a calibration technique for CuAAC kinetics, or more generally, “click” kinetics in membrane functionalization, and underscores the utility of the more rigorous PDE model.

Solid lines in Figure 5 depict simulation results of the membrane functionalization process solving the DNCuAC concentration predicted by the ink preparation equilibrium model, eqs 1–10, and the calibrated reaction-diffusion PDEs, eqs 28b and 28c. As seen in Figure 5, the prediction accuracy increases with copper concentration (from C to A) and aligns better with the overall functionalization progress, as measured by the time-series conversion data informed by FTIR. Experiments with initial Cu(II) concentrations at 100 mM (Figure 5A) and 50 mM (Figure 5B) correspond to an average model prediction error of less than 6.3%. However, Figure 5C cautions against predictions at a lower initial Cu(II) concentration (i.e., 10 mM), with an average prediction error of 9.3%. This could be attributed to the higher relative errors in the DNCuAC concentration predictions from our ink preparation equilibrium model for low-Cu(II) ink solution where an unmodeled reaction or intermediate (e.g., MCuPIC) may dominate. Therefore, joint with the initial DNCuAC concentration inferred from the ink preparation equilibrium model, the reaction-diffusion PDE model—encompassing first-principle diffusion and dual first-order reaction between azide and DNCuAC—presents strong predictive capability for the general functionalization process.

4.3. Reaction-Diffusion PDE Model Outperforms the Instantaneous Reaction Model in Membrane Functionalization.

Plotting the conversion of azide moieties in the membrane against the DNCuAC concentration helps visualize the comparison between functionalization models without variability from the ink preparation model. FTIR-informed azide conversion data from Figure 5 are transferred to Figure 7 via the ink preparation equilibrium model DNCuAC predictions, facilitating a direct comparison between the two membrane functionalization models. Specifically, Figure 7A displays the predictions of the reaction-diffusion PDE model, and Figure 7B displays model predictions for the instantaneous reaction model, solved as a function of time and DNCuAC concentrations through algebraic manipulation of eq 26. For Figure 7A,B, the experimental data remain the same, yet the model predictions are clearly unique. Both predictions of azide conversion include ± 3 s bands to account for operational deviations.

This analysis explicitly demonstrates the superior accuracy of the reaction-diffusion PDE model in predicting azide conversion across all processing times and DNCuAC

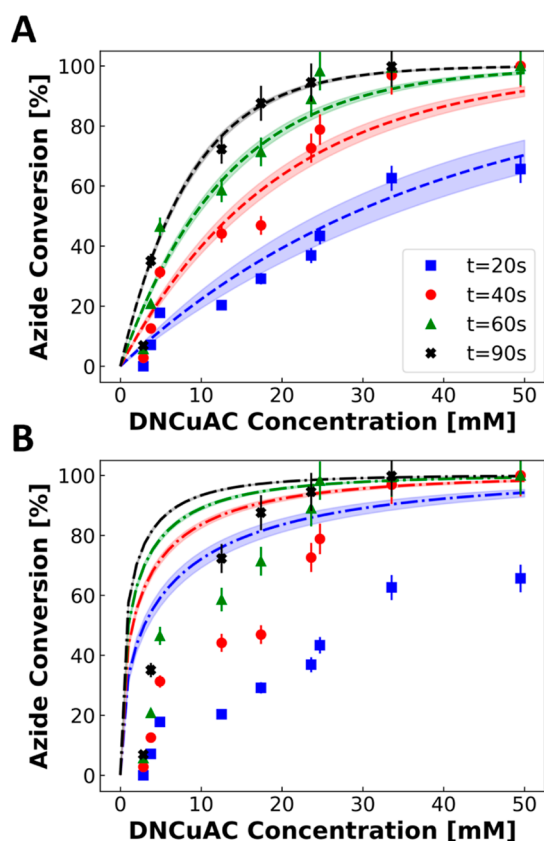


Figure 7. Azide conversion predicted by the (A) reaction-diffusion PDE model ($D = 6.3 \mu\text{m}^2\cdot\text{s}^{-1}$ and $k_a = 1.69 \text{ M}^{-1}\cdot\text{s}^{-1}$) and (B) instantaneous reaction model ($D = 0.9 \mu\text{m}^2\cdot\text{s}^{-1}$) at varied DNCuAC concentrations and exposure times. The data points displayed utilize DNCuAC concentration predicted from the ink preparation equilibrium model based on ink formulations. The dashed lines correspond to model predictions of azide conversion with $\pm 3\%$ deviation bands.

concentrations below 50 mM. Consistent with Figure 5, the PDE model is inaccurate at DNCuAC concentrations below 5 mM. In Figure 7B, experimental points deviating from the trend signify that the process does not meet the limit of an instantaneous reaction, i.e., $D_a = 51.27$ does not sufficiently satisfy $D_a \gg 1$. For DNCuAC concentrations greater than 25 mM, data points for 60 and 90 s begin to overlap with the instantaneous reaction trends as the limit of conversion is reached. For these high conversion points, the linear regression method of Hoffman et al.²¹ can be used to estimate D in eq 27; however, it may be challenging to generate large amounts of data in this region via Edisonian exploration. In contrast, the PDE model can be used to analyze all of the data. These high conversion points also suggest that even though the CuAAC reaction may not be instantaneous, it is capable of achieving high azide conversion with longer processing times. The instantaneous reaction model, though it fails to predict conversion for the general functionalization processes ($D_a = 51.27$), provides a physical lower bound of D and depicts the most ideal scenario with kinetic improvements. As highlighted in Section 2.4, instantaneous reactions are preferred in functionalization because they provide sharp boundaries for different chemical functionalities, and the discrete chemical distributions are easy for the on-demand design and characterization of new membranes. Improvements in kinetic rates can

be achieved with the addition of ligands to further increase the CuAAC reaction rate in achieving larger D_a .^{66,67} Furthermore, the instantaneous reaction model, with its analytic solution form, could provide preliminary insights into evaluating the investment in ligands against cost savings from copper usage. Supporting Information Figure S5 compares the average conversion for the two model predictions at different membrane thicknesses, which informs that the instantaneous reaction model proves most effective for 2 to 3 μm membranes with processing times exceeding 60 s and shows decreased reliability with thinner membranes.

4.4. Quantifying Copper Usage and Functionalization Time Trade-Offs. Drawing a horizontal line in Figure 7A shows that there are multiple experimental conditions with different DNCuAC concentrations to achieve a target conversion rate with varying processing times. Figure 8 directly visualizes the relationships between the ink preparation and the corresponding processing time. In Figure 8A, azide conversions are calculated successively through our ink preparation model and reaction-diffusion PDE model, with varying initial copper concentration $[\text{Cu}^{2+}]_0$ and processing time. The ink formulation was fixed at $[\text{Cu}^{2+}/\text{AA}/\text{DMA}] = [1:1:2]$, which was identified as the optimal ink formulation providing DNCuAC concentration with minimal copper usage. In this regard, dashed lines in Figure 8A represent the solution points for optimization problem eqs 28a–28c, solved with different conversion requirements η^* and processing times t_{end} . For example, achieving a 90% conversion in 40 s for a 15 μm membrane necessitates a copper concentration greater than 94 mM. For a specific conversion requirement, there are opportunities to lower copper concentration with a longer processing time—a finding consistent with our insights from Figure 7A. Navigating between these lines, higher η^* tends to require higher copper concentration or longer processing time. These trade-off relationships between the minimal copper usage and the corresponding processing time for functionalization provide valuable design criteria for the membrane functionalization process. It is noteworthy that the results in this section are simulated for 15 μm membranes. Experimental validation of the optimal ink formulation and processing time with 10, 5, and 0.4 μm membranes is provided in Supporting Information Figure S6.

Figure 8B,C presents the operational space for different ink formulations. Similar to Figure 8A, azide conversions are predicted with varying initial copper concentration $[\text{Cu}^{2+}]_0$, processing time, and at several ink formulations. 90% azide conversion lines are selected and plotted together to determine optimal ink formulations. In Figure 8B, reducing $[\text{DMA}/\text{Cu}^{2+}]$ ratio from 2:1 results in increased copper usage and processing time, leading to worse manufacturing process design. Increasing the $[\text{DMA}/\text{Cu}^{2+}]$ ratio, on the other hand, reaches a limit where additional DMA offers no improvement in azide conversion. Similarly, Figure 8C highlights a $[\text{AA}/\text{Cu}^{2+}]$ ratio limit at 0.6. Compared to copper, the economic and environmental cost of the AA is modest, hence we adjusted the $[\text{AA}/\text{Cu}^{2+}]$ ratio to avoid precipitation of copper in ink preparation. Rather than a consecutive mapping from ink formulation to azide conversion, Figure S7 illustrates the DNCuAC concentration and Cu(II)/DNCuAC conversion evaluated for different ink formulations. Consistent with Figure 8B,C, there are clear limits where additional DMA and AA provide no benefit in DNCuAC concentration or Cu(II)/

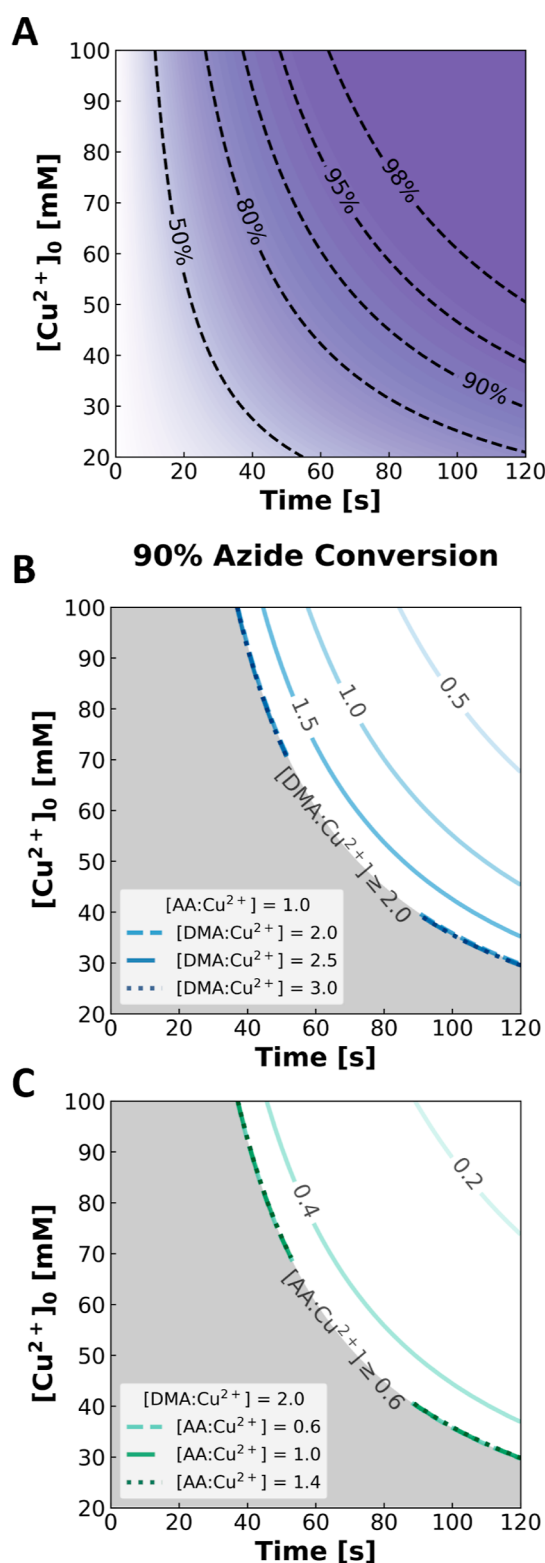


Figure 8. Relating azide conversion to ink formulations. Panel (A) corresponds to contours evaluating azide conversion at varying copper usage and functionalization time with a fixed $[\text{Cu}^{2+}/\text{AA}/\text{DMA}]$ ratio of $[1:1:2]$. Panels (B) and (C) represent 90% azide conversion contour moving from optimal ink formulation with (B) changing $[\text{DMA}/\text{Cu}^{2+}]$ ratio $[\text{Cu}^{2+}/\text{AA}/\text{DMA}] = [1:1:N]$ and (C) changing $[\text{AA}/\text{Cu}^{2+}]$ ratio $[\text{Cu}^{2+}/\text{AA}/\text{DMA}] = [1:N:2]$. The gray areas on the bottom left denoting the physical infeasible region, i.e., 90% azide conversion, cannot be achieved.

DNCuAC conversion. These findings are consistent with Hoffman et al.,²¹ who reported that ink solutions containing 30 mM Cu(II) and propiolic acid, prepared at 0.3 Cu/alkyne and 0.5 Cu/alkyne, led to similar azide conversion profiles. This suggests the common presence of thresholds for the ink component ratios in CuAAC reactions. Using our optimal ink formulation $[\text{Cu}^{2+}/\text{AA}/\text{DMA}] = [1:1:2]$, Supporting Information Figure S8 presents a linear relationship between copper usage and DNCuAC concentration. Also, Cu(II)/DNCuAC conversion is always greater than 49%, approaching 50% at high copper concentrations. These results further echo the optimal design criteria from Figure 8A using the ink formulation of $[\text{Cu}^{2+}/\text{AA}/\text{DMA}] = [1:1:2]$.

Table 4 shows the optimal operational conditions that minimize the total copper investment in a continuous

Table 4. Optimal Copper Usage and Functionalization Time for Continuous Membrane Manufacturing Obtained from Optimization Problem 30

η^* [%]	$[\text{Cu}^{2+}]_0$ [mM]	t_{end} [s]
90	32	112
95	40	116
98	60	102

multifunctional membrane manufacturing process. For a specific azide conversion target η^* , the objective function, eq 30, was evaluated at the trade-off lines in Figure 8A to identify the unique condition that contributes to the minimal total copper investment, eq 29. For example, the optimal operational conditions to achieve 90% functionalization, which requires 32 mM Cu(II) with 112 s processing time, enable at least 20% reduction in total copper investment compared to merely adopting the ink formulation from previous experimental studies.²¹ Although full copper recovery and reuse were assumed to derive the optimization objective in eq 30, due to the weight structure, we expected the optimal solutions to also hold when accounting for different levels of copper losses. Such an evaluation as eq 29 formulates a compact utility function for the multiobjective decision-making problem in process design facing conflicting factors. This approach offers a perspective that balances the trade-offs between ink solution preparation and functionalization bath length design, providing crucial information for techno-economic assessment and LCA before scaling-up the manufacturing process of multifunctional membranes and implementation in novel separation systems,⁶⁸ especially when considering upgrades from conventional copolymer-based membrane systems.

4.5. Functionalization Improves Membrane Performance. Finally, we performed diafiltration experiments^{33,69,70} to demonstrate that optimized membrane functionalization improves separation performance. Figure 9 examines the changes in MgCl_2 rejection with respect to the concentration of the feed solution. The blue triangles correspond to diafiltration experiments conducted on a 10 μm -thick azide-functionalized parent membrane. For diafiltration experiments conducted under similar experimental and operating conditions, the rejection of Mg^{2+} by membranes functionalized at optimal conditions (i.e., azide conversion $\geq 98\%$) increases (Figure 9, orange squares). Here, the dimethylamine groups are protonated and introduce positive charges throughout the membrane. The functionalization process leads to distinct changes in the performance of membrane samples, thereby

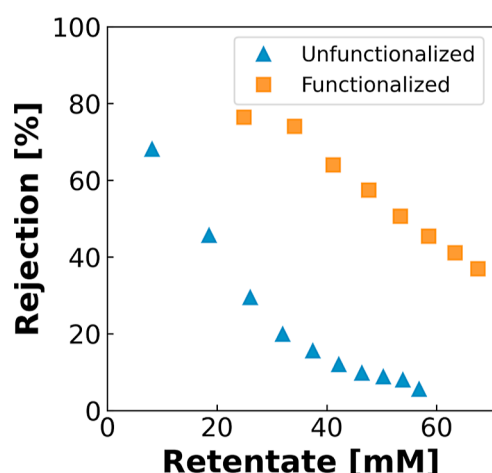


Figure 9. MgCl_2 rejection for unfunctionalized (i.e., azide) membranes in blue triangles and optimally functionalized (i.e., dimethylamine) membranes in orange squares. The membrane samples have thicknesses of 10 μm . The data points correspond to diafiltration experiments run with a 1 mM MgCl_2 feed and 80 mM MgCl_2 diafiltrate.

confirming the successful conversion of the azide group and providing a proof of concept for optimizing structured and multifunctional membrane manufacturing. The [Supporting Information](#) further describes the diafiltration experiments.

5. CONCLUSIONS

This study develops a comprehensive ink optimization framework to scale up novel multifunctional membrane manufacturing. Our data science workflow enabled rigorous and efficient model identification of the functionalization mechanisms by leveraging various experimental techniques. The proposed equilibrium model maximizes the availability of catalytically active species while minimizing the cost of ink formulations. Computational tools accelerated the calibration and simulation of the reaction-diffusion PDEs, enabling more accurate predictions of general membrane functionalization progress compared to the instantaneous reaction model. Our multifaceted investigation contributes to the fundamental understanding of membrane functionalization processes, resulting in the two-stage model that quantitatively mapped ink formulation to functionalization progress control. An optimal ink formulation was identified at $[\text{Cu}^{2+}/\text{AA}/\text{DMA}] = [1:1:2]$, which minimizes copper concentration and discloses trade-offs between copper usage and functionalization time for achieving the desired azide conversions. Moreover, we prototyped a continuous multifunctional membrane manufacturing process with conserved functionalization baths, which links conflicting design factors to total copper investment, leaving a single solution to the decision-making problem for functional membrane commercialization. The presented models and analyses serve as a foundation for manufacturing process design (e.g., reaction selection, ink optimization, and functionalization equipment design), guiding future research in the pursuit of sustainable and efficient novel membrane production processes.

There are several future research directions. Toward fundamental scientific research, optimal experiment design³⁶ can refine model identification that elucidates more details of the process. For example, it is crucial to note that the current ink preparation mechanistic model, while performing well in

predicting azide conversions when coupled with the PDE model, lacks direct evidence to quantify the concentrations of DNCuAC. The modest 10% uncertainty in the estimated diffusion coefficient from FTIR-informed conversion can impact the functionalized boundary profile, leading to a loss of resolution in functional features. Extending the data science workflow to leverage parameter precision criteria from MBDoE could help in the design of more informative experimental protocols and improve the characterization and modeling of functionalization processes. Beyond membrane science, the FTIR-monitored membrane functionalization process, with elaborated and well-formulated mechanisms, may serve as a new platform for investigating and calibrating “click” kinetics for broader applications, e.g., biomedical⁷¹ and pharmaceutical^{72,73} synthesis. Toward practical applications, the optimization framework can be extended to design continuous manufacturing processes for complicated structured membranes. Current analysis and verification experiments were conducted on the laboratory scale. While the presented results provide significant insights, further validation at the pilot scale is necessary to ensure applicability and robustness in large-scale manufacturing processes. The conversion target presented in the optimization problem allows for convenient verification through FTIR, capturing a complete azide conversion over a representative thickness of $\sim 4 \mu\text{m}$. This can easily be replaced with other conversion considerations to evaluate the azide concentration profile in practical functionalization processes, e.g., using a spatial average concentration of the functionalized group over a designed chemical thickness. Furthermore, the versatility associated with our problem formulation can accommodate the precise additive manufacturing of membranes with a vast array of pore wall chemical pattern designs. For example, patterns tailored to different functionalities can prevent fouling and selectively capture multiple components throughout the membrane simultaneously.¹⁹ The development of multifunctional membranes, such as Janus membranes, requires investigation into the introduction of subsequent inks subject to different initial and boundary conditions within the PDEs, expanding the overall capabilities of membranes. Ultimately, this work provides upscaling protocols for novel chemical pattern design in membrane structures, facilitating their applications and advancing sustainable and fit-for-purpose separations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c11749>.

Detailed model reformulation and nondimensionalization, AIC derivation, additional experiments including Cu(II)-AA pH experiments for model identifiability analysis, Cu(II)-DMA complex UV-vis experiments for model selection, and verification of functionalization results (PDF)

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Notes

The authors declare no competing financial interest.

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NOMENCLATURE

Abbreviations

AA	ascorbic acid
AIC	Akaike information criterion
CuAAC	copper(I)-catalyzed azide–alkyne cycloaddition
DI	deionized
DMA	3-dimethylamino-1-propyne
DNCuAC	dinuclear copper–alkyne π complex
FIM	Fisher information matrix
FTIR	Fourier-transform infrared
LCA	life cycle assessment
MBDoE	model-based design of experiments
MCuPIC	mononuclear copper π complex
MPI	multiple processing interface
NF	nanofiltration
PDE	partial differential equations
TFE	trifluoroethanol
WLS	weighted least-squares

Data science methods

θ	model parameter estimate
$\Sigma_{\hat{y}}$	model prediction covariance matrix

θ	model parameter vector
M	Fisher information matrix
Q	sensitivity matrix
S	eigenvectors of the FIM
u	constant/controlled experimental condition vector
$V_{\hat{\theta}}$	parameter covariance matrix
x	experimental condition variable vector
y	experimental observation vector
$\chi^2_{(\cdot)}$	χ^2 distribution
ΔAIC	differences in AIC
ΔAIC_c	differences in corrected AIC
ϵ	experimental observation error
η_{bc}	Bartlett correction factor
$\hat{y}$	model prediction variable
Λ	eigenvalues of FIM
$\mathbb{R}$	real numbers
$\mathcal{L}(\cdot)$	likelihood function
σ	standard deviation
Σ_y	covariance matrix of measurement errors
$f(\cdot)$	differential or algebraic equations
$F_{(\cdot)}$	F distribution
k	model complexity
N	total number of experimental observations
n_r	number of experimental observations for response r
p	number of model parameters
$p(\cdot)$	probability distribution function
R	number of response types
y	experimental observation variable

Model variables and parameters

C	vector assembles multicomponent molar concentration of the component being determined by UV–vis, M
k	vector assembles products of multicomponent molar absorptivity and length of the UV–vis light path through the sample, M^{-1}
w	weight vector for ink formulation decision variables, dimensionless
x	reactive ink formulation vector, M
ϵ_{λ}	molar absorptivity or molar extinction coefficient at wavelength λ , $M^{-1}\cdot cm^{-1}$
η^*	azide conversion target, dimensionless
η	conversion of azide, dimensionless
γ	inverse of the depth that the evanescent wave of the FTIR laser penetrates into the sample, μm^{-1}
τ	dimensionless variable for t
ϵ	emissivity of the sample, $M^{-1}\cdot \mu m^{-1}$
ζ	dimensionless variable for z
A	intensity of the FTIR absorbance signal, dimensionless
A_{λ}	UV–vis absorbance at wavelength λ , dimensionless
A_m	membrane area exposed to the reactive solution, μm^2
b	length of the UV–vis light path through the sample, cm
C	molar concentration of the component being determined by UV–vis, M
c_A	dimensionless variable for C_A
c_D	dimensionless variable for C_D
C_{A0}	concentration of parent azide, M
C_A	concentration of azide, M
C_{D0}	DNCuAC concentration at equilibrium, M
C_D	concentration of active reactant DNCuAC, M
D	diffusion coefficient of the active reactant in the membrane, $\mu m^2\cdot s^{-1}$
D_a	Damköhler number, dimensionless

k_a	overall reaction rate constant for general functionalization, $M^{-1}\cdot s^{-1}$
K_e	overall equilibrium constant for Cu(II) to Cu(I) by AA, M^2
k_R	functionalization reaction rate constant, $M^{-1}\cdot s^{-1}$
K_{a1}	equilibrium constant for hydrolysis of ascorbic acid, M
K_{a2}	equilibrium constant for hydrolysis of DMA, M
K_{e2}	equilibrium constant for Cu(I) formation, M
K_{i2}	equilibrium constant for $CuDH^+$ formation, M^{-1}
K_{i3}	equilibrium constant for Cu(I) formation from $CuDH^+$, M^2
K_{II}	equilibrium constant for DNCuAC formation, M^{-2}
K_{s2}	equilibrium constant of Cu(II)–DMA complex $Cu(DMA)^{2+}$ formation, M^{-1}
K_{s3}	equilibrium constant of Cu(II)–DMA complex $Cu(DMA)_2^{2+}$ formation, M^{-2}
K_w	water dissociation constant, M^2
L'	ink penetration depth, μm
L	membrane thickness, μm
l	characteristic length associated with diffusion, μm
L_{bath}	length of functionalization bath, m
t	ink penetrating time or time elapsed since the ink solution contacts the membrane surface, s
t_{end}	functionalization processing time, s
U_{Cu}^{2+}	total copper investment
w	weight for WLS
z	diffusion depth of active reactant DNCuAC, μm

Other symbols

$[\cdot]$	aqueous concentration of a species, M
$[\cdot]_0$	initial concentration of a species, M
$v_{[\cdot]}$	log 10-transformed concentration

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